



Department of Engineering Cybernetics

TTK4147 Real-time Systems

11 – Distributed Systems - Architectures

05.11.2025

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Today

- About «Systems»
- How to describe a (distributed) system?
(Architectures)
 - System models
 - Software architectures
 - System architectures
 - Client – server architecture
 - Peer-to-peer architecture
 - Overlay network
- Communication on physical network
- Architectures and real time properties

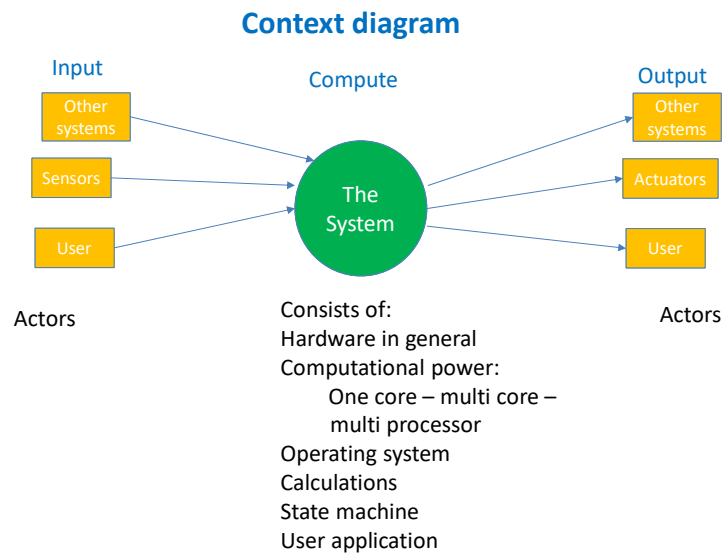
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Systems, in general



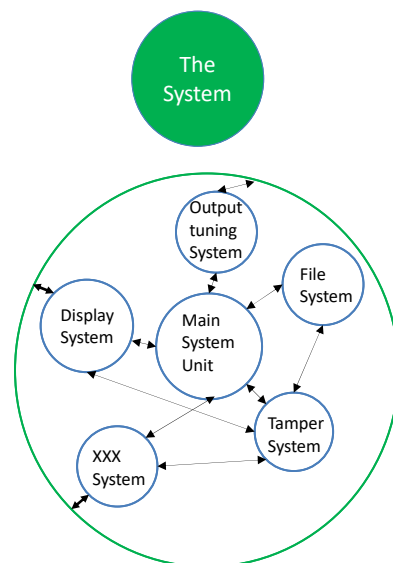
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What is a / one System?



- Systems into a system
- Brother – Sister Systems

Edward Yourdon: Modern Structured Analysis

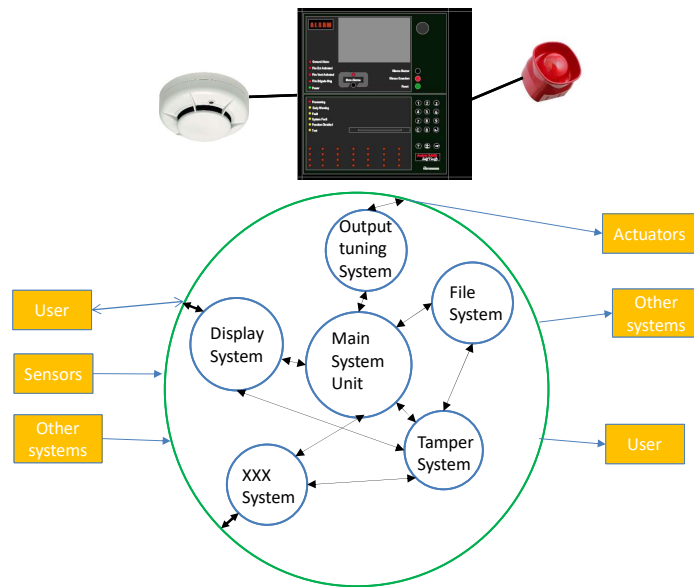
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Local system



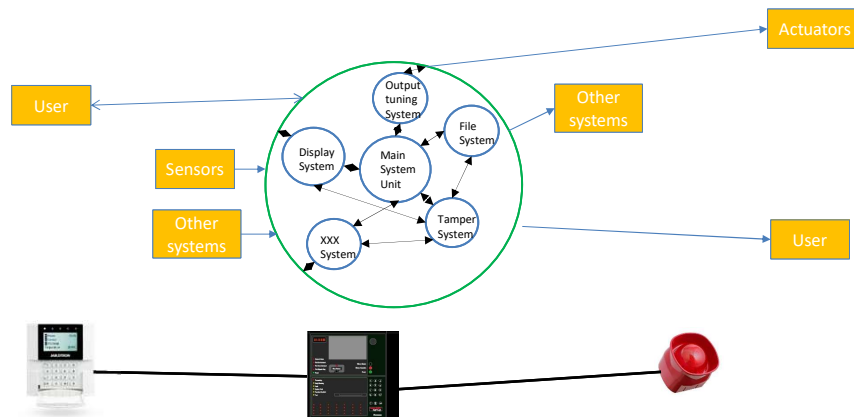
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Geographically distribution



What if the distances to some actors increase?

Due to physical circumstances, some components must be distributed.

You may be able to compensate with longer cables

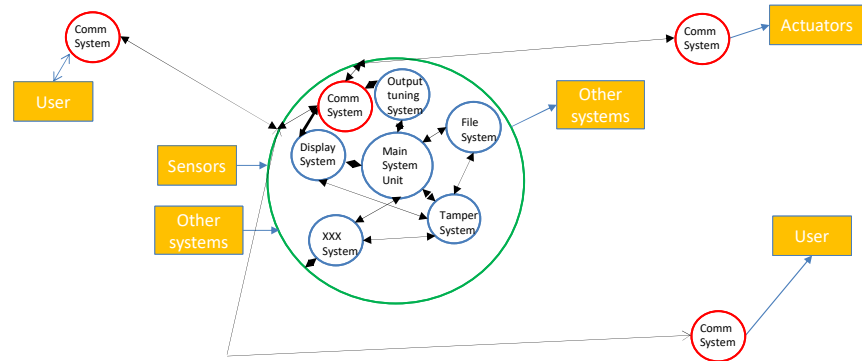
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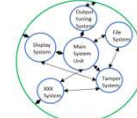
Geographically distributed system 1



At a given distance, you will implement some kind of communication link into your system

This communication link may be an information exchange bottleneck.

Previous system

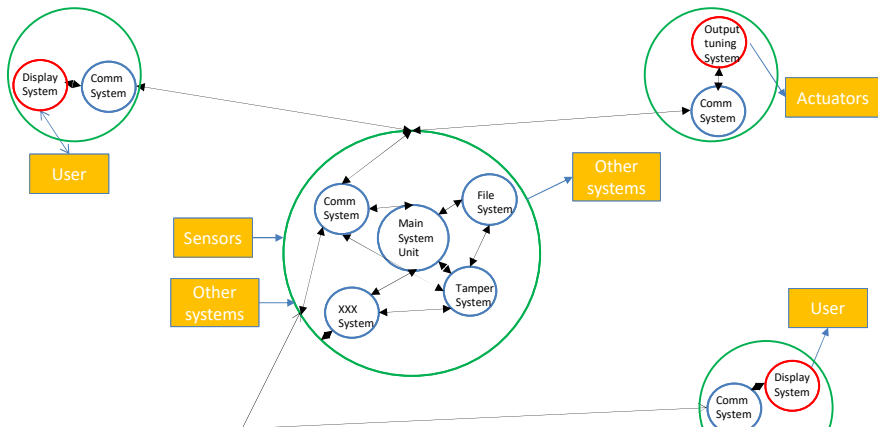


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Geographically distributed system 2



Possible to move some of the data processing from the central part of the system to the distributed parts of the system, e.g. to reduce the communication (decrease bottlenecks), increase the robustness.

Will often use standardized interface between systems to keep the possibility to change vendors.

Previous system



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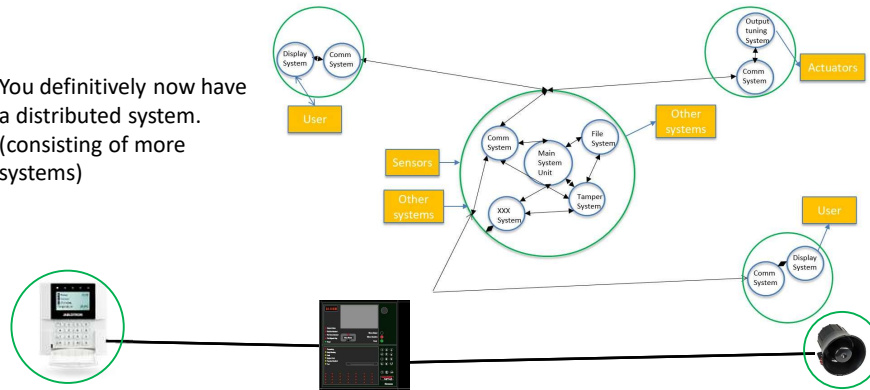
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Geographically distributed system 3

You definitively now have a distributed system.
(consisting of more systems)



Such system of smart distributed systems have become more and more common, mainly because of availability of powerful yet cheap microprocessors (PCs, workstations, PDAs, embedded systems, etc.) and continuing advances in communication technology.

Bear in mind the foregoing slides when we go through the rest of the lecture

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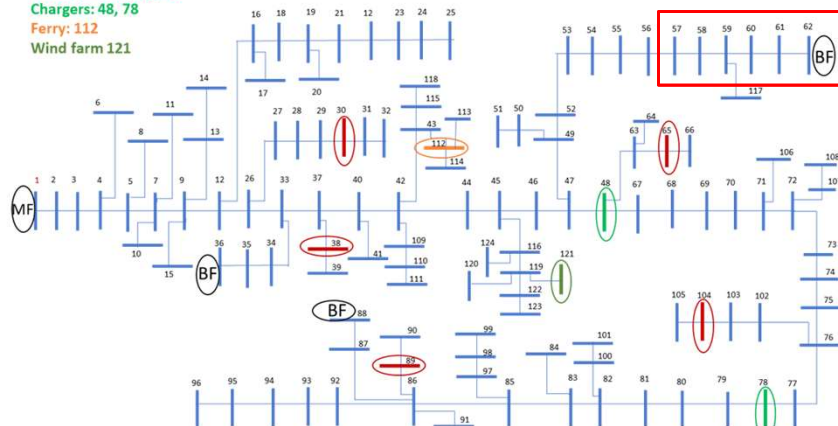


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Why focus on system design?

Motivation from the electric power distribution grid

main feeder: 1
Backup feeders: 36, 62, 88
LEC: 30, 38, 65, 89, 105
Chargers: 48, 78
Ferry: 112
Wind farm 121



Schematic presentation of a power grid

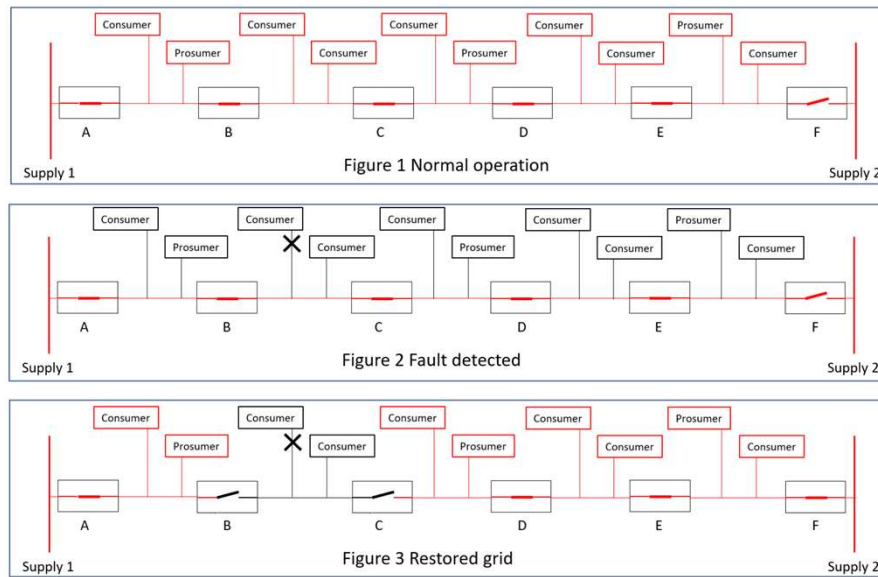
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Small section of electric grid with self healing



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Electric grid with SH, where to locate the controllers?

Possible location of control system (architectures):

- a) Distributed, one in each node

(A) (B) (C) (D) (E) (F)

- b) Local centralized, one common for the involved nodes, one higher up

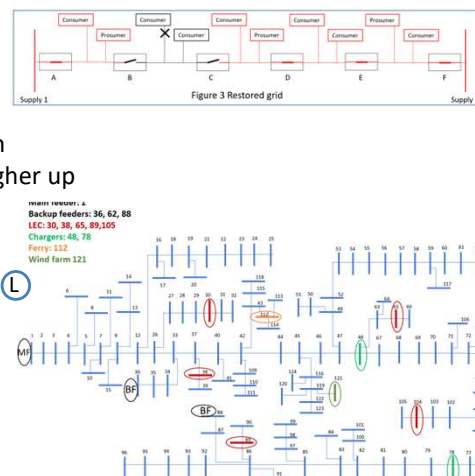
(M) (M) (M)

(A) (B) (C) (D) (E) (F) (G) (H) (I) (J) (K) (L)

- c) Centralized, one common for all nodes in the complete grid

(M)

(A) (B) (C) (D) (E) (F) (G) (H) (I) (J) (K) (L) (M) (N) (O) (P) (Q) (R) (S) (T) (U) (V) (W) (X) (Y) (Z)



The architecture influences on response time, robustness, maintainability ...

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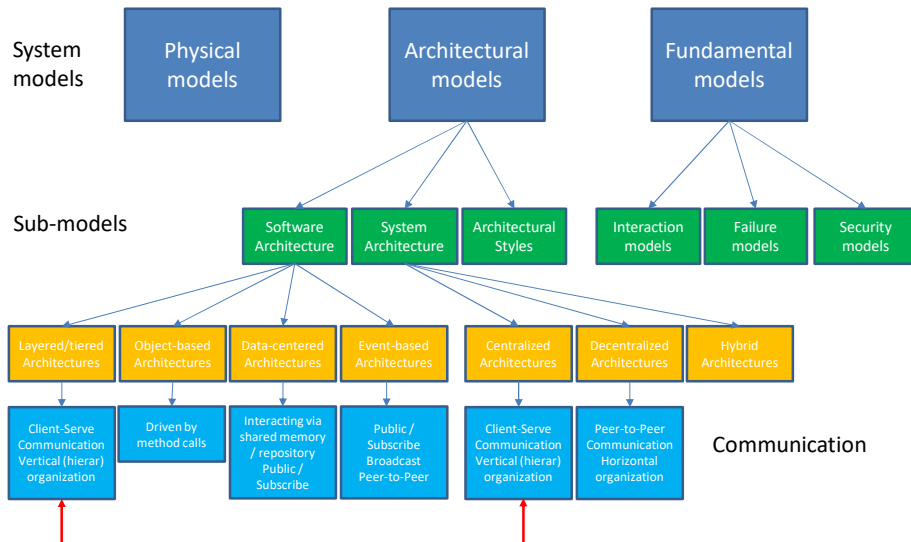
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How to describe a (distributed) system? (an overview)

Hierarchy:



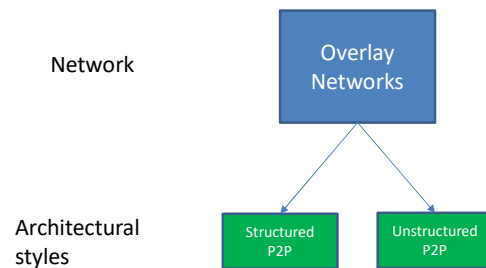
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How to describe a (distributed) system continue?



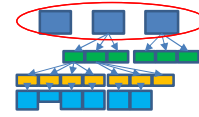
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System Models



- Three complementary ways to describe and discuss design of distributed systems



- The System models illustrate / describe common properties and **design choices** for distributed systems.

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Three types of system models



- Physical models
 - Physical models consider the types of computers and devices that form a system and their interconnectivity.
 - No details of specific technologies
- Architectural models
 - Describe a system in terms of the computational and communication tasks performed by its computational elements and connecting network
 - Client-server** and **peer-to-peer** are two of the most commonly used forms of architectural model for distributed systems.
- Fundamental models
 - Take an abstract perspective in order to describe solutions to individual issues faced by most distributed systems

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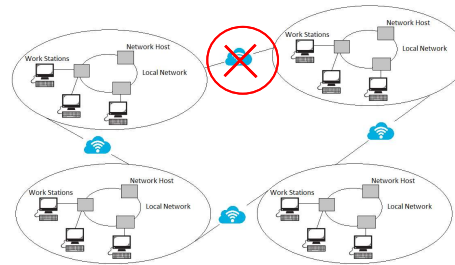
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System model 1: Physical models



- Physical models consider the types of computers and devices that form a system and their interconnectivity.
- No details of specific technologies.
Physical models: Capture the hardware composition of a system in terms of computers and other devices and their interconnecting network.



A debatable detail in the picture?



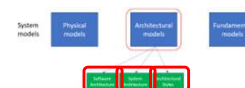
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Architectural Models



The architecture of a system is its structure in terms of separately specified components and their interrelationships.

Architectural models define:

- The main components of the system, what their roles are and how they interact (**software architecture**).
- How they are deployed in an underlying network of computers (**system architecture**).
- A particular way to configure a collection of components* and connectors** (**architectural style**, the structure of the software based on which the design is created).

* Component: A module with well-defined interfaces; reusable, replaceable.

** Connector: A communication link between modules.

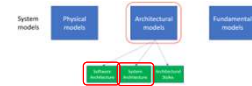
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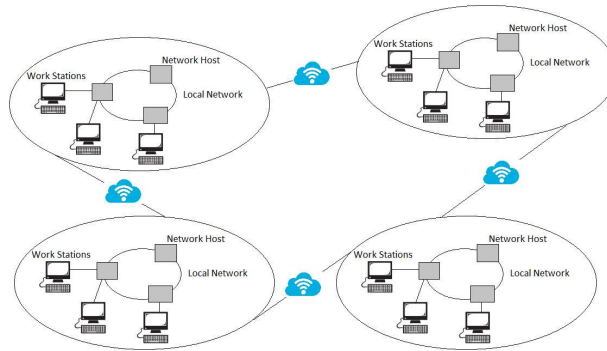
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Architectural Models continue



Architecture model is concerned with the **placement of its parts**, how components are mapped to underlying network and the relationship between them.



Software architecture
System architecture

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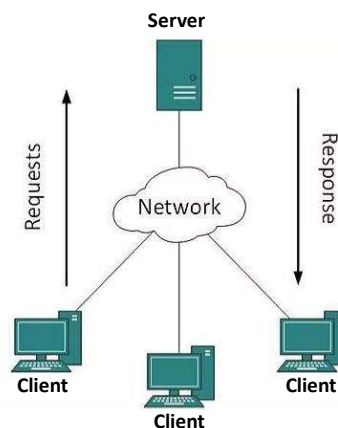
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Client-Server Architecture

Often used in distributed systems.

- The architecture has two major components, the client and the server.
- The Server is (in principle) where all the processing, computing and data handling is happening.
-  The clients can make requests to the Server, and the Server will respond accordingly.
- In particular, client processes interact with individual server processes in potentially separate host computers in order to access the shared resources that they manage.
- The figure illustrates the simple structure in which clients make requests to a server, which the server response on.



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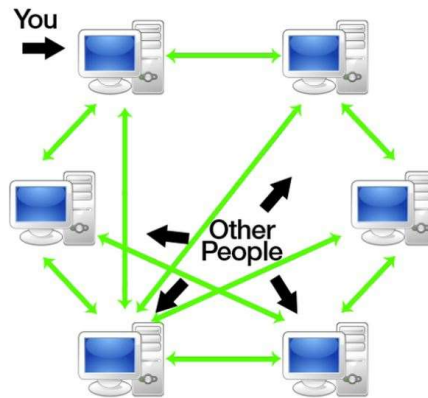
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Peer-to-Peer (P2P) Architecture

A style also often used in distributed systems.

The general idea behind peer to peer style is where there is no central control in a distributed system.

Each node (in general, referred to as a Peer) can in principle communicate with every other.



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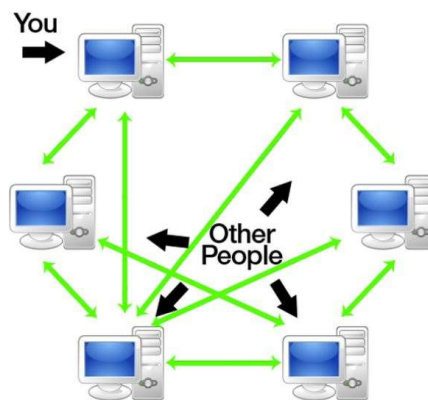


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Peer-to-Peer (P2P) Architecture, *continue*

The basic idea is that each node can either be a client or a server at a given time.

If the node is requesting something, it can be known as a client, and if some node is providing something, it can be known as a server.



Changing roles as server / client; from Linux:
Loadable-, stackable modules.

(Hierarchic organized: Individual modules serve as libraries when they are referenced by client modules higher up in the hierarchy, and as clients when they reference modules further down.)

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Comparison Client / Server – P2P Architectures

BASIS FOR COMPARISON	CLIENT-SERVER	PEER-TO-PEER
Basic	There is a specific server and specific clients connected to the server.	Clients and server are not distinguished; each node can act as client and server.
Service	The client request for service and server respond with the service.	Each node can request for services and can also provide the services.
Focus	Sharing the information.	Connectivity.
Data	The data is stored in a centralized server.	Each peer has its own data.
Server	When several clients request for services simultaneously, the server can be a bottleneck.	As the services are provided by several servers distributed in the peer-to-peer system, a server will not be a bottleneck.
Expense	The client-server are expensive to implement.	Peer-to-peer are less expensive to implement.
Stability	Client-Server is more stable and scalable.	Peer-to-Peer suffers if the number of peers increases in the system.

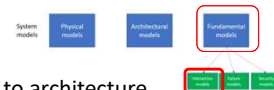
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Fundamental Models



Gives a formal description of the properties that are common to architecture models. All system models have some common fundamental properties.

3 fundamental models:

1. Interaction models:

Interaction models describe how the different elements in an object model interact with each other.

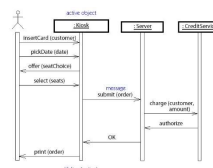
Processes can communicate with messages and coordinate via synchronization for synchronization / ordering of activities.

Challenges, in general:

- The message delays are often of considerable duration
- Coordination between processes is limited by lack of global clock.

Examples of Interaction models:

- Use case diagram
- Sequence diagram
- Activity diagram
- Collaboration diagram



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Fundamental Models continue



3 fundamental models (continue):

2. Failure models:

The failure model attempts to give a precise specification of the faults that can effect:

- Processes (e.g. process halts, doesn't meet deadline in hard RT)
- Communication channels. (e.g. message vanish, get incorrect check sum)

3. Security models:

- Define and classify the forms attacks against processes and communication channels may take.
- Basis for the analysis of threats to a system and for the design of resistible systems.
- Introduces the concept of a secure channel

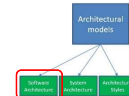
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Software Architecture



Software architectures common for distributed systems:

1. Layered architectures
2. Object-based architectures
3. Data-centered architectures
4. Event-based architectures

The focus on these architectures are on how they communicate

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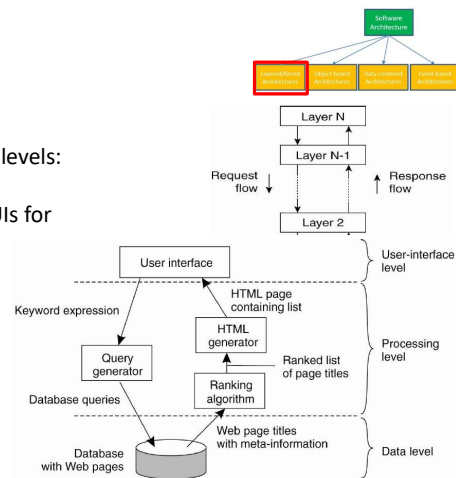


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Layered Architectures

E.g. for Client-Server systems:
This layered architecture consists of three levels:

- User-interface level:** It usually uses GUIs for interacting with end users.
- Processing level:** It is concerned with data processing applications, which is the core functionality.
- Data level:** interacts with data base or file system. Data usually is persistent; exists even if no client is accessing it. It is concerned with file or database system.



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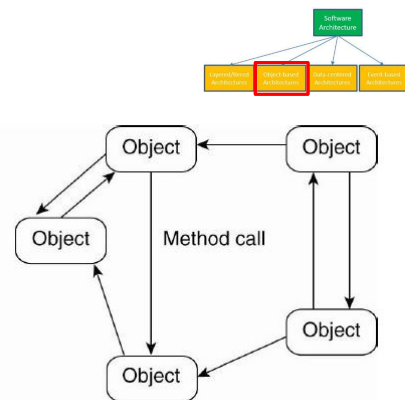
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Object-based architectures

- Based on loosely coupled arrangement of components with no specific architecture.
- Each of the components are referred to as objects, which can interact with other objects through a direct method call.

Examples:

- Java Remote Method Invocation (Java RMI)
- RESTful **API** (REST* API)



* Representational State Transfer (REST) is a software architecture initially created as a guideline to manage communication on a complex network like the internet

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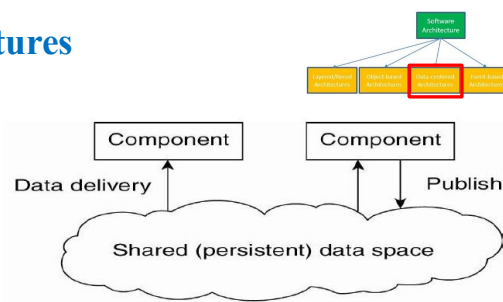
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Data-centered architectures

- Main purpose is data access and data update.
- Processes interact by reading and modifying data in some shared repository in either active or passive manner.
 - a) Traditional data base (passive): responds to requests
 - b) Blackboard system (active): clients solve problems collaboratively; system updates clients when information changes.
- Communication via event propagation.
E.g. in distributed systems, Publish-Subscribe is often used.



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What is Publish-Subscribe?

Definition:

Publish-Subscribe messaging is an asynchronous communication method in which messages are exchanged between applications without knowing the identity of the sender or recipient.

Overview:

Four core concepts make up the model:

- **Topic** – An intermediary channel that maintains a list of subscribers to relay messages which are received from publishers
- **Message** – Serialized messages sent to a topic by a publisher which has no knowledge of the subscribers
- **Publisher** – The application that publishes a message to a topic
- **Subscriber** – An application that registers itself with the desired topic in order to receive the appropriate messages

Primary advantages:

- Loose coupling
- Scalability.

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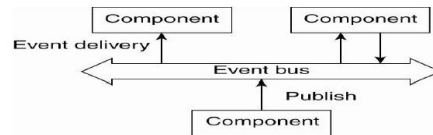
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Event-based architectures



- All communication in this kind of a system is event-driven. When an event is generated, it will be communicated. Then, “everyone” are notified about the occurrence of the event.
- Used with several communication styles:
 - a) Publish-Subscribe
 - b) Broadcast
 - c) Peer-to-Peer
- Often used in systems with loosely coupled components.



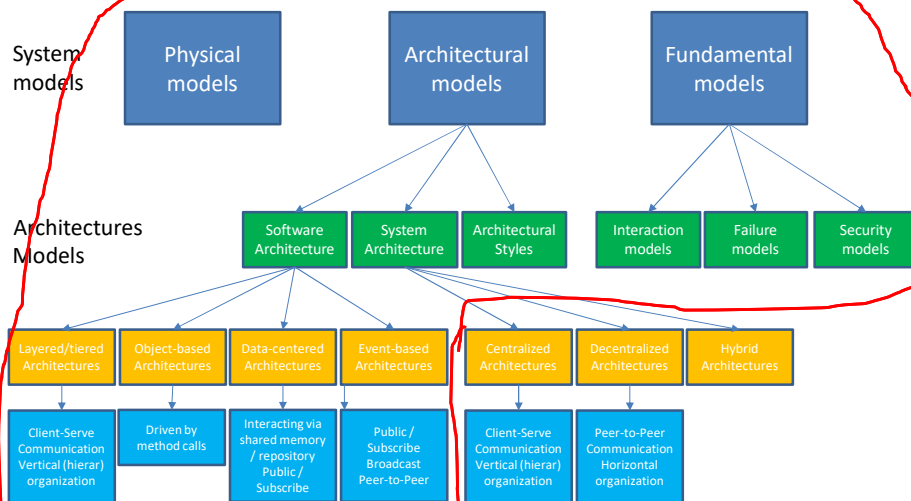
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How to describe a (distributed) system?



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Other aspect to consider concerning SW architectures

The degree of coupling between tasks / subsystems:
Loosely- or Tightly coupled systems.

Loosely coupled systems (from Wikipedia):

- A loosely coupled system is one in which each of its components has, or makes use of, little or no knowledge of the definitions of other separate components.
- Components in a loosely coupled system can be replaced with alternative implementations that provide the same services, e.g. from other vendors.
- Components in a loosely coupled system are less constrained to the same platform, language, operating system, or build environment.

(Properties we are seeking)

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Tightly coupled system, common characteristics

- The style has low scalability
- The system has often a monolithic structure (not modular)
- It has low delays 
- Data rate in tightly coupled systems is high 
- Throughput is high in tightly coupled systems 
- Security is (often) high in tightly coupled systems. 



Why?

→ Wanted characteristics that often results in a tightly coupled system 😊

→ Positive characteristics in Loosely CS are often missing in Tightly CS and opposite.

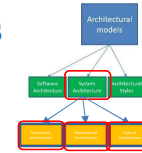
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System Architectures for Distributed Systems



The architecture for distributed systems can be

- **Centralized**
characterized by traditional client-server structure.
 - **Vertical** (or hierarchical) organization of communication and control paths (as in layered software architectures)
 - Logical separation of functions into client (requesting process) and server (responder)
- **Decentralized:**
characterized by peer-to-peer.
 - **Horizontal** (flat) rather than hierarchical communication and control
 - Communication paths are less structured; symmetric functionality.
- **Hybrid:**
A combination of elements using Client Server and Peer-to-Peer communication.

Classification of a system as centralized or decentralized refers (here) to communication and control organization, primarily.

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Why to distribute a systems?

In our terms, to distribute a system means to divide the complete task into subtasks running on different processors. Normally the reasons are:

- The system consists of commercial of the shelf modules.
- To increase the computational power.
- Because the physical task to solve is geographically distributed



The distribution of a system will result in distribution and duplication of hardware and software and often addition of communication modules.

However, to distribute a system raise other fundamental challenges

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How / what to distribute in Distributed Systems

Centralized, Decentralized, Hybrid systems should be evaluated in a broader scope than communication and control organization.

In a multi node system, two important design issues have to be discussed:

Where to store which data?

- Centralizing of data increases the consistence, but also increase the access time, which may reduce the system efficiency dramatically and gives single point of failure.
- Decentralizing of data may from time-to-time result in data- and mode-inconsistencies in the system.



Examples?

E.g. Access control to houses (Mode:public open / key needed)

Where to process which routines?

- A trend is to move I/O data conditioning to the front-end (edge computing).
- Requires computational (and electric) power at the front-end
- Edge computing often minimize the need for communication
- Can increase autonomy and robustness against communication failure

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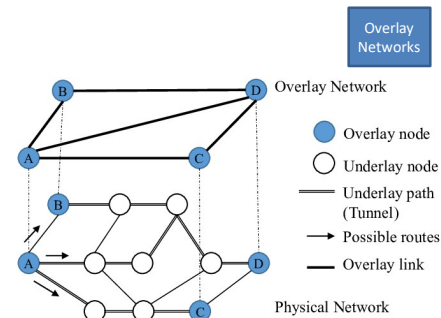
Overlay Networks

- Overlay networks are logical or *virtual* networks, built on top of a physical network.
- A link between two nodes in the overlay may consist of several physical links.
- Messages in the overlay are sent to logical addresses, not physical (IP) addresses.
- An overlay network may be **structured** (nodes and content are connected according to some design that simplifies later lookups) or **unstructured** (content is assigned to nodes without regard to the (overlay) network topology).

Various approaches used to resolve logical addresses to physical.



Examples? From QNX? Use of Global Name Service (GNS) and GNS manager



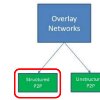
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Structured P2P Systems



In structured peer-to-peer networks the overlay is organized into a specific topology, and the protocol ensures that any node can efficiently search the network for a file/resource, even if the resource is extremely rare.

A common approach is to use a distributed hash table (DHT) to organize the nodes.

Traditional hash functions convert a key to a hash value, which can be used as an index into a hash table. The keys are unique, each represents an object to store in the table. The hash function value is used to insert an object in the hash table and to retrieve it.

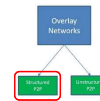
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Structured P2P Systems *continue*



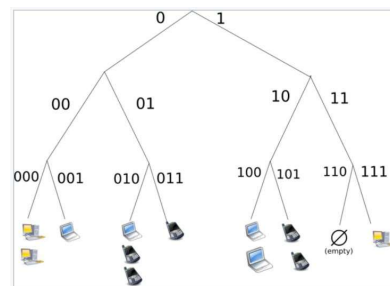
In a DHT, both data objects and nodes are each assigned a key which hashes to a single random number.

A mapping function assigns data objects to nodes, based on the hash function value.

A lookup, also based on hash function value, returns the network address of the node that stores the requested object.

In DHT the information (key / value pair) is distributed on some few nodes and duplicated on other nodes

The system handles addition and removal of nodes with less effort



Overlay network diagram for a **structured P2P network**, using a **distributed hash table (DHT)** to identify and locate nodes/resources

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Structured P2P Systems *continue*

From Wikipedia:

In a DHT, both data objects and nodes are each assigned a key which hashes to a random number.

A mapping function assigns objects to nodes, based on the hash function value.

A lookup, also based on hash function value, returns the network address of the node that stores the requested object.

In DHT the information (key / value pair) is distributed on some few nodes and duplicated on other nodes. The structure of a DHT can be decomposed into several main components. [12][13] The foundation is an abstract keyspace, such as the set of 160-bit strings. A keyspace partitioning scheme splits ownership of this keyspace among the participating nodes. An overlay network then connects the nodes, allowing them to find the owner of any given key in the keyspace.

Once these components are in place, a typical use of the DHT for storage and retrieval might proceed as follows. Suppose the keyspace is the set of 160-bit strings. To index a file with given filename and data in the DHT, the SHA-1 hash of filename is generated, producing a 160-bit key k , and a message $\text{put}(k, \text{data})$ is sent to any node participating in the DHT. The message is forwarded from node to node through the overlay network until it reaches the single node responsible for key k as specified by the keyspace partitioning. That node then stores the key and the data. Any other client can then retrieve the contents of the file by again hashing filename to produce k and asking any DHT node to find the data associated with k with a message $\text{get}(k)$. The message will again be routed through the overlay to the node responsible for k , which will reply with the stored data.

The keyspace partitioning and overlay network components are described below with the goal of capturing the principal ideas common to most DHTs; many designs differ in the details.



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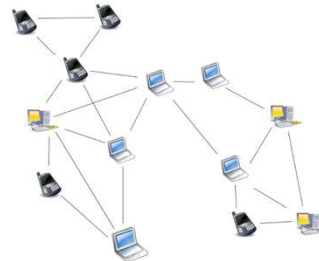
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Unstructured P2P Systems

- Unstructured P2P organizes the overlay network as a random graph.
- Each node knows about a subset of nodes, its neighbors.
(Neighbors are chosen in different ways: physically close nodes, nodes that joined at about the same time, etc.)
- Data items are randomly mapped to some node in the system and lookup is random.



Overlay network diagram for an **unstructured P2P network**, illustrating the ad hoc nature of the connections between nodes

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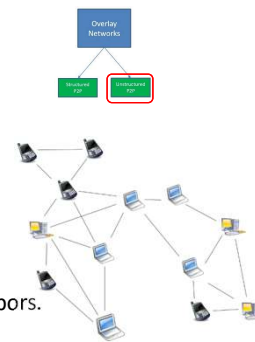


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Unstructured P2P Systems continue

How to locate a Data Object by using Flooding:

- Send a request to all known neighbors
- If not found, neighbors forward the request to their neighbors.
- Works well in small to medium sized networks, doesn't scale well.
- "Time-to-live counter" can be used to control number of hops.
- As the network grows, flood searches become very CPU- and time-consuming



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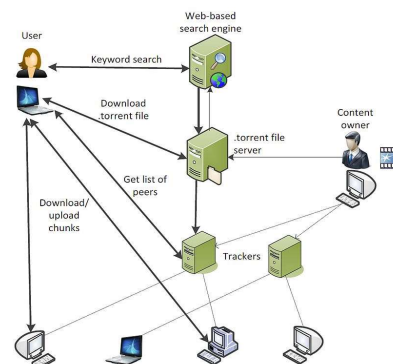


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Hybrid P2P Systems

Hybrid systems are often based on both client-server architectures and p2p networks.

- A well known example is Bittorrent. The torrent search engines provide a client server architecture, where the trackers provide a structured p2p overlay. The rest of nodes, which are also known as leechers* and seeders**, become the unstructured overlay of the network, allowing it to scale itself as needed and further.



* "leechers": Users who are downloading the files

** "seeders": Users who have the complete file and are sharing it

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Summary on Structured and Unstructured P2P Systems

	Unstructured P2P	Structured P2P.
Overlay construction	High flexibility	Low flexibility
Resources	Indexed locally	Indexed remotely on a distributed hash table
Query messages	Broadcast or random walk	Unicast
Content location	Best effort	Guaranteed
Performance	Unpredictable	Predictable bounds
Overhead	High	Relatively low
Object types	Mutable, with many complex attributes	Immutable. with few simple attributes
Peer churn & failure	Supports high failure rates	Supports moderate failure rates
Applicable environments	Small-scale or highly dynamic, e.g. mobile P2P	Large-scale & relatively stable, e.g. desktop file sharing
Examples	Gnutella, LimeWire, KaZaA, Bit Torrent	Chord, CAN. Pastry, eMule, Bit Torrent

"Churn"—when large numbers of peers are frequently joining and leaving the network

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Communication on Physical network

Physical (geographically) distribution of systems requires communication.

When designing a distributed system, there are some factors to consider concerning the physical network communication:

- Wired or wireless communication
- Infrastructure* to use:
 - Private
 - Public shared
 - Private public
- Protocol to use
- The deterministic properties of the network

***Infrastructure** (here) refers to the composite hardware, software, network resources and services required for the existence, operation and management of electronic communication

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Wired or Wireless communication

Wired (copper / fiber)

- + Wired communication is (more) difficult to tamper
- + Often possible to support power in the same cable as the communication uses
- + Reliable communication
- + Well proven technology

Wireless

- + Wireless communication saves installation cost of nodes and cables
- + Can be used in installations where (communication) cables are impossible (e.g. retrofits)
- + Can be used to moveable / wearable equipment
- + Contributes to miniaturization of equipment
- For battery-powered systems:
 - Must consider range of operation vs. frequency of communication vs. amount of information vs. power consumption
- The connectivity may be a problem (predictability?)

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Hybrid Wired and Wireless communication



- Lots of wireless sensors with short range communication → save energy
- “Impossible” to cable the sensors
- Communicate with a wired base station

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Communication infrastructures

The different communication infrastructures to use in a distributed system influence on

- Development / installation time ("Time to operation")
- Cost (development / operation)
- Availability of existing infrastructures
- Security

Examples of communication infrastructures:

- Private comm. infrastructure
- Public shared comm. infrastructure
- Private public comm. infrastructure

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Communication infrastructures continue

Private comm. infrastructure:

- Reasons for to choose: To assure connectivity. (If cost calculation recommend so)
- Implemented as
 - Own cable or own developed wireless modules
 - Built on commercial off-the-shelfs (COTS) components (e.g. LoRa equipment, Private 5G)
- + Security through proprietary solutions
- + Possible to tune the power consumption, tailor made
- May be expensive to extend and to maintain

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Communication infrastructures continue

Public shared comm. infrastructure:

- + “Ready to use”
- + Other are running the infrastructure and doing the maintenance.
- May be tampered / unsecured
- (Normally) Have to pay an ever-lasting fee
- May have varying capacity
- E.g. 2G, 4G, 5G, Public internet, LoRa

Private public comm. infrastructure

- + Using Public shared infrastructure, tunneling via SW, e.g. Virtual Private Network (VPN)
- + Increase security
- Increase SW footprint and computational load

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Communication protocols to use

Factors influencing selection of communication protocol:

- Proprietary protocols:
 - Special tuned,
 - Energy efficient,
 - Communication efficiency,
 - Simpler drivers (HW/SW)
- Standardized protocols:
 - Gives interoperability,
 - Ready-to-use SW stack and HW drivers available.
- SW footprint → size of communication stack
- Requirements from other communication partners
- Reliability
- Security
- System requirements (e.g. Real-Time)

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Architectures and Real Time Properties

Although Real Time properties are more about predictability than speed, the efficiency of a distributed system (and by that often the ability being predictive) depends on the architecture.

Examples of architectural issues that may influence on the predictability:

- Client-Server Architecture, is the computational power of the server well balanced to the number of clients?
- P2P Architecture, is the number of nodes well balanced to the (un-)structured network?
- What does the interaction model tell about the message delays?
- What does the failure model tell about the robustness of the system?
- Is the chosen SW architecture well fitted for your application
- Is the choice between centralized / decentralized architecture optimal for your application?
- Is the communication in the distributed system predictable?

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Also use

Blackboard → TTK4147 Sanntidssystemer (2025 HØST) →
Learning materials → Questions

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Exercise

2023-8

Decide whether the statements below are true or false.

	True	False
Throughput is higher in tightly coupled systems than in loosely coupled systems.....	X	
The general idea behind client-server architecture is that there is no central control in a distributed system.....		X
Publish-Subscribe messaging is an asynchronous communication method in which messages are exchanged between applications without knowing the identity of the sender or recipient.....	X	
A common approach in unorganized peer-to-peer systems is to use a distributed hash table (DHT) to organize the nodes.....		X

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Exercise

2020-1i

What message passing method could be used when applications share messages, but all applications should remain incognito?

Publish-Subscribe

Why is this method often used?

It is easy to scale, and the couplings between the applications are looser

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Exercise

2023-4

Name four basic concepts that make up the publish-subscribe model.

Publisher, subscriber, topic, message

2021-18

How can one locate data in an unstructured peer-to-peer system??

Flooding: Send a request to all known neighbors. If not found, neighbors forward the request to their neighbors

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Exercise

2019-12

Give some advantages by using wired (copper / fiber) and wireless communication

Wired (copper / fiber):

- Wired communication is (more) difficult to tamper
- Often possible to support power in the same cable as the communication uses
- Reliable communication
- Well proven technology

Wireless:

- Wireless communication saves cost and installation cost of cables
- Can be used in installations where (communication) cables are impossible (e.g. retrofits)
- Can be used to moveable / wearable equipment
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